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Studierichting: Informatica

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$$\begin{aligned}\int \frac{1}{x^4 - 1} dx &= \int \frac{1}{(x^2 - 1)(x^2 + 1)} dx = \int \frac{1}{(x - 1)(x + 1)(x^2 + 1)} dx \\&\rightarrow \frac{1}{(x - 1)(x + 1)(x^2 + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{Cx + D}{x^2 + 1} \\&\Rightarrow A = \frac{1}{(1 + 1)(1 + 1)} = \frac{1}{4} \\&\Rightarrow B = \frac{1}{(-1 - 1)((-1)^2 + 1)} = -\frac{1}{4} \\&\Rightarrow C(i) + D = \frac{1}{(i - 1)(i + 1)} = -\frac{1}{2} + 0i \\&\Rightarrow D = -\frac{1}{2}, C = 0 \\&= \frac{1}{4} \int \frac{1}{x - 1} dx - \frac{1}{4} \int \frac{1}{x + 1} dx - \frac{1}{2} \int \frac{1}{x^2 + 1} dx \\&= \frac{1}{4} \ln |x - 1| - \frac{1}{4} \ln |x + 1| - \frac{1}{2} \operatorname{Bgtan} x + c\end{aligned}$$

References